Reading & Interpreting Statistical Diagrams

Exam code: 0607

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Reading & interpreting statistical diagrams

How do I interpret statistical diagrams?

- Read and **understand** the initial sentences describing the **situation** (context)
  - **Underline** important words if necessary
- Look for any **keys** that may help you to understand the diagram
  - For example
    - 1 unit represents 20 people
    - Year 10 is the **solid** line, Year 11 is the **dotted** line
    - Class A is **shaded**, class B is **striped**
- **Read** the **titles** of diagrams and all **axes labels**
  - A graph for '**new** students at a school' is different to '**all** students at the school'
  - A **frequency** axis that **starts at 50** is different to one starting at 0
- Understand which **units** are being used
  - Individual lengths may be in **centimetres** but total length may be in **metres**
  - Populations may be measured in **thousands**
- Look out for any **extreme values** (outliers / anomalies)
  - One month's temperature might be unusually high
    - Was it a heat wave or a recording error?

How do I draw conclusions from diagrams?

- Look for **overall trends** in the diagram
  - Prices **increase** year on year
  - The temperature **peaks** in June
- Use **numbers** from the graphs
- Refer to any **changes**
  - The **steepness** (gradient) of graph may change
- Write in **full sentences** that copy the **exact wording** from the question

Help me understand

Stuck on something?

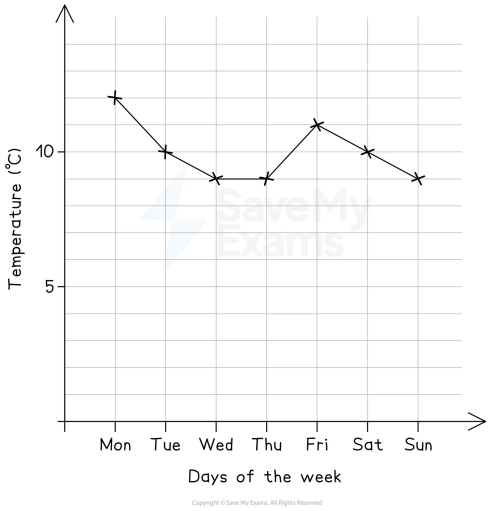
Ask questions and get help with exactly what you're revising.

Got it

- You may need to calculate the **mode**, **median**, **mean** or **range** to support any explanations
- Understand why drawing conclusions may **not** be **suitable**
  - The data set may be **too small** to be representative
  - The data set may be **biased**
  - Consider the **scope** of the data
    - e.g. Data for January to March cannot be used to predict August

Worked Example

The diagram below shows the temperature of a garden in the UK, recorded at 7am on each day of a particular week in March.



(a) Describe the change in temperature over the first four days.

Answer:

The trend shows a decrease in the first three days, then a constant temperature
Find numbers from the graph to use in your answer
Refer to the steepness of the change

The temperature decreases from 12°C on Sunday to 9°C on Wednesday
The decrease is steeper over the first two days
There is then a constant temperature of 9°C on both Wednesday and Thursday

(b) A gardener claims that, based on the graph, Monday must have experienced the highest temperature that week.

Give a reason as to why this might not be true.

Answer:

The temperatures on the graph are at 7am each day
The maximum temperature may have been after 7am, on a day that was not Monday

(c) A journalist wants to use the data shown to claim that the average temperature that week was below 10°C.
The mean of the temperatures shown is 10°C.

Which type of average would you suggest they use? Explain your answer.

Answer:

The mean of 10°C does not support the claim that the average temperature is below 10°C
Try calculating the mode instead

12, 10, 9, 9, 11, 10, 9

The most frequent number is 9

The modal temperature is 9°C
9°C < 10°C so using the mode would help the journalist's claim

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